

VALDEZ WSO, AK, USA

WMO: 702750

Lat: 61.130N

Lon: 146.352W

Elev: 95

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 94-05

WBAN: 26442

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	7.5	-14.2	2.6	8.9	-10.7	3.1	9.3	32.6	17.9	29.3	19.2	13.6	60	0.690

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	70.7	56.5	67.8	56.0	65.0	55.0	58.9	67.9	57.4	65.5	56.1	62.7	9.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
54.8	64.1	60.0	53.8	61.7	58.5	52.8	59.4	57.4	25.5	68.4	24.6	65.9	23.8	63.0	63.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	19.6	16.5	1.0	79.1	4.1	3.6	-1.9	81.7	-4.3	83.8	-6.6	85.9	-9.6	88.5
			-0.9	60.8	3.7	1.5	-3.6	61.9	-5.8	62.8	-7.9	63.6	-10.6	64.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	23.7	27.1	30.5	38.9	47.3	54.4	55.8	54.6	47.9	38.8	29.7	25.8	(6)
	DBStd	12.95	8.69	7.54	5.99	5.03	4.60	4.07	3.34	3.81	4.08	5.28	6.34	7.46	(7)
	HDD50	4303	816	642	605	334	108	5	0	4	84	347	609	749	(8)
	HDD65	9274	1281	1062	1070	782	550	320	287	324	513	812	1059	1214	(9)
	CDD50	507	0	0	0	2	23	136	179	146	21	0	0	0	(10)
	CDD65	2	0	0	0	0	0	1	1	0	0	0	0	0	(11)
	CDH74	39	0	0	0	0	0	12	16	11	0	0	0	0	(12)
	CDH80	2	0	0	0	0	0	1	1	0	0	0	0	0	(13)
(14) Wind	WSAvg	6.0	7.8	6.4	7.4	5.2	5.9	6.0	4.7	4.2	4.2	5.9	7.2	7.3	(14)
(15) Precipitation	PrecAvg	65.9	5.9	5.6	4.4	3.5	3.0	2.8	3.7	6.4	8.7	8.3	5.7	7.2	(15)
	PrecMax	93.3	15.2	14.2	10.0	8.1	6.7	6.1	9.0	18.2	16.7	15.4	20.6	17.3	(16)
	PrecMin	34.8	0.0	0.6	0.8	0.6	0.7	0.1	0.2	0.4	0.7	0.3	0.4	1.3	(17)
	PrecStd	12.8	3.8	3.4	2.6	2.0	1.4	1.4	1.9	3.9	3.9	4.0	4.3	4.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	39.3	46.1	62.1	68.8	75.6	76.0	75.4	69.5	53.5	44.0	40.7	(19)
		MCWB	35.4	34.1	36.4	48.2	50.0	57.6	58.4	57.5	53.4	42.8	39.9	34.7	(20)
	2%	DB	37.0	37.5	42.4	54.8	63.9	70.5	69.2	69.2	62.0	50.2	41.5	38.0	(21)
		MCWB	33.6	33.5	35.1	44.0	49.6	56.5	58.0	57.1	50.4	41.7	39.2	35.0	(22)
	5%	DB	35.1	36.2	40.0	50.1	59.6	67.1	66.2	66.0	57.9	48.0	39.3	35.9	(23)
		MCWB	32.7	33.1	34.2	41.3	48.0	55.6	57.1	56.4	49.0	41.5	35.4	33.5	(24)
	10%	DB	33.8	35.0	38.0	47.2	56.6	63.6	63.3	63.0	54.3	46.0	37.2	34.3	(25)
		MCWB	32.1	32.7	33.3	39.9	47.0	54.0	56.0	55.1	48.2	41.1	34.4	32.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	36.4	39.0	48.5	52.8	60.3	60.3	60.1	54.1	48.0	42.2	38.3	(27)
		MCDB	40.6	37.4	42.3	60.8	63.7	71.7	70.0	68.8	63.0	48.6	42.7	38.9	(28)
	2%	WB	34.7	35.3	37.0	44.8	50.8	58.1	58.8	58.5	52.4	45.7	40.4	36.0	(29)
		MCDB	35.8	36.2	40.0	53.6	60.3	67.8	67.7	66.7	58.5	47.0	41.2	36.8	(30)
	5%	WB	33.5	34.3	35.5	42.3	49.6	56.3	57.4	57.0	51.0	44.0	37.1	34.3	(31)
		MCDB	34.3	35.2	38.2	48.3	57.7	64.9	65.2	64.4	54.9	45.6	38.1	35.0	(32)
	10%	WB	32.5	33.3	34.4	40.8	48.2	54.6	56.2	55.5	50.0	42.6	35.1	33.3	(33)
		MCDB	33.5	34.2	36.7	45.7	55.1	62.3	62.2	61.3	53.1	44.4	36.4	34.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.3	10.0	12.2	14.1	14.8	12.7	13.9	11.6	8.7	7.1	7.2	(35)
		MCDBR	8.2	8.7	12.3	18.1	21.5	22.8	20.1	21.4	19.3	11.6	8.5	7.7	(36)
		MCWBR	6.1	6.4	8.2	10.3	11.2	10.9	10.2	11.0	10.3	7.0	5.6	5.5	(37)
	5% WB	MCDBR	6.8	6.5	9.7	15.7	17.8	20.5	17.6	18.9	13.7	8.5	6.0	6.0	(38)
		MCWBR	5.7	5.2	6.8	9.4	10.0	10.1	9.5	10.4	8.2	6.4	4.2	5.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.259	0.318	0.325	0.386	0.402	0.404	0.404	0.401	0.343	0.319	0.273	0.251	(40)
	taud		2.006	2.026	2.065	1.940	2.039	2.165	2.230	2.235	2.362	2.188	1.984	1.895	(41)
	Ebn at Noon		137	196	245	249	257	260	257	246	243	198	124	79	(42)
	Edh at Noon		14	26	35	48	47	42	39	36	26	22	14	9	(43)
(44) All-Sky Solar Radiation	RadAvg		90	313	815	1390	1734	1794	1607	1222	787	360	129	49	(44)
	RadStd		10	43	78	107	121	101	98	115	101	40	20	5	(45)

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page